DNA Damage - Blerd

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Data Set Up

```
# 1.
library(tidyverse)
```

```
Warning: package 'ggplot2' was built under R version 4.4.3

── Attaching core tidyverse packages ──────────────────────── tidyverse 2.0.0 ──
✔ dplyr     1.1.4     ✔ readr     2.1.5
✔ forcats   1.0.0     ✔ stringr   1.5.1
✔ ggplot2   4.0.1     ✔ tibble    3.2.1
✔ lubridate 1.9.4     ✔ tidyr     1.3.1
✔ purrr     1.0.4
── Conflicts ────────────────────────────────────────── tidyverse_conflicts() ──
✖ dplyr::filter() masks stats::filter()
✖ dplyr::lag()    masks stats::lag()
ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to b
```

```
library(car)
```

```
Loading required package: carData

Attaching package: 'car'

The following object is masked from 'package:dplyr':

    recode

The following object is masked from 'package:purrr':

    some
```

```
library(gt)
library(modelsummary)
library(emmeans)
```

```
Welcome to emmeans.
Caution: You lose important information if you filter this package's results.
See '? untidy'
```

```
library(modelbased)
```

```
Warning: package 'modelbased' was built under R version 4.4.3
```

```
library(multcomp)
```

```
Loading required package: mvtnorm

Attaching package: 'mvtnorm'

The following object is masked from 'package:modelbased':

    standardize

Loading required package: survival
Loading required package: TH.data
Loading required package: MASS

Attaching package: 'MASS'

The following object is masked from 'package:dplyr':

    select


Attaching package: 'TH.data'

The following object is masked from 'package:MASS':

    geyser
```

```
library(ggpubr)
library(cowplot)
```

```

Attaching package: 'cowplot'

The following object is masked from 'package:ggpubr':

    get_legend

The following object is masked from 'package:gt':

    as_gtable

The following object is masked from 'package:lubridate':

    stamp
```

```
library(lme4)
```

```
Warning: package 'lme4' was built under R version 4.4.3

Loading required package: Matrix

Attaching package: 'Matrix'

The following objects are masked from 'package:tidyr':

    expand, pack, unpack
```

```
library(lmerTest)
```

```

Attaching package: 'lmerTest'

The following object is masked from 'package:lme4':

    lmer

The following object is masked from 'package:stats':

    step
```

```
library(outliers)
library(DHARMa)
```

```
This is DHARMa 0.4.7. For overview type '?DHARMa'. For recent changes, type news(package
```

```
library(statmod)
library(readr)
library(dplyr)
library(EnvStats)
```

```

Attaching package: 'EnvStats'

The following object is masked from 'package:Matrix':

    print

The following object is masked from 'package:MASS':

    boxcox

The following object is masked from 'package:car':

    qqPlot

The following objects are masked from 'package:stats':

    predict, predict.lm

The following object is masked from 'package:base':

    print.default
```

```
library(performance)
```

```
Warning: package 'performance' was built under R version 4.4.3
```

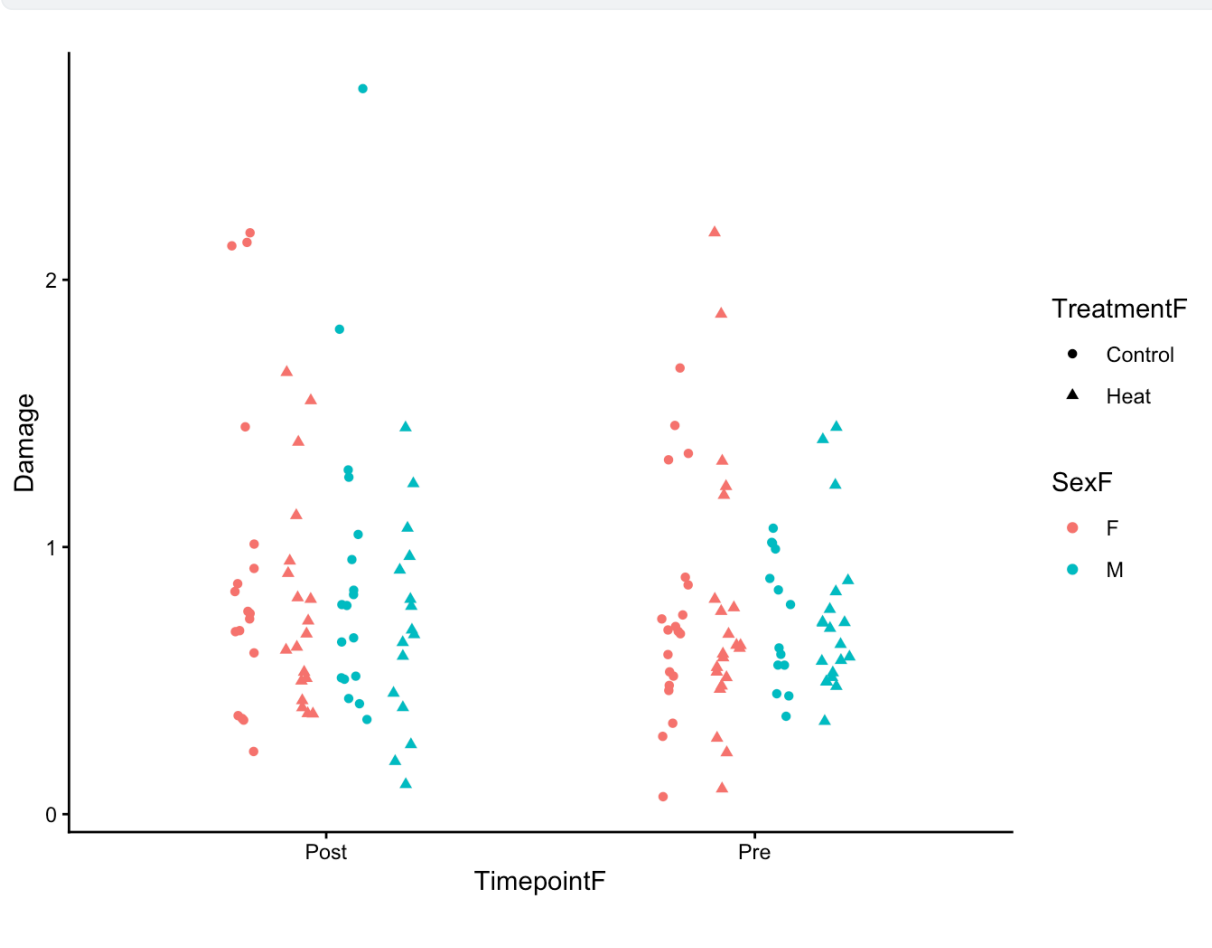
```
library(see)
library(patchwork)
```

```

Attaching package: 'patchwork'

The following object is masked from 'package:cowplot':

    align_plots

The following object is masked from 'package:MASS':

    area
```

```
# 2.
setwd("/Users/ben/Desktop/Blerd/Data Analysis")

# 3. Load your data
datum = read.csv("Blerd.csv")
datumlong = read.csv("BlerdData_long.csv")

# 4. Ensure categorical variables are properly set
datum$SexF = as.factor (datum$Sex)
datum$TreatmentF = as.factor(datum$Treatment)
datum$TimeF = as.factor(datum$Time)
datum$IncubatorF = as.factor (datum$Incubator)
datum$SideF = as.factor (datum$Side)
datum$PostDamage = as.numeric (datum$PostDamage)
datum$PreDamage = as.numeric (datum$PreDamage)

datumlong$SexF = as.factor (datumlong$Sex)
datumlong$TreatmentF = as.factor(datumlong$Treatment)
datumlong$TimepointF = as.factor(datumlong$Timepoint)
datumlong$Damage = as.numeric (datumlong$Damage)

# 5. Subset data so there are no dead individuals included
datum1 = datum %>%
  filter(!Died %in% c("Died", "PIT.Death"))

datumlong1 = datumlong %>%
  filter(!Died %in% c("Died", "PIT.Death"))
datumlong1 = datumlong1[-47, ]

# 6. Take out any of the blank cells that cause problems when graphing
datum2 = datum1 %>%
  filter(PostDamage != "", PostDamage != "Dead", AvgMaxTemp != "", AvgMaxTemp != "Dead")
datum2$TreatmentF = as.factor(datum2$Treatment)

datumlong2 = datumlong1 %>%
  filter(Damage != "", Damage != "Dead")
```

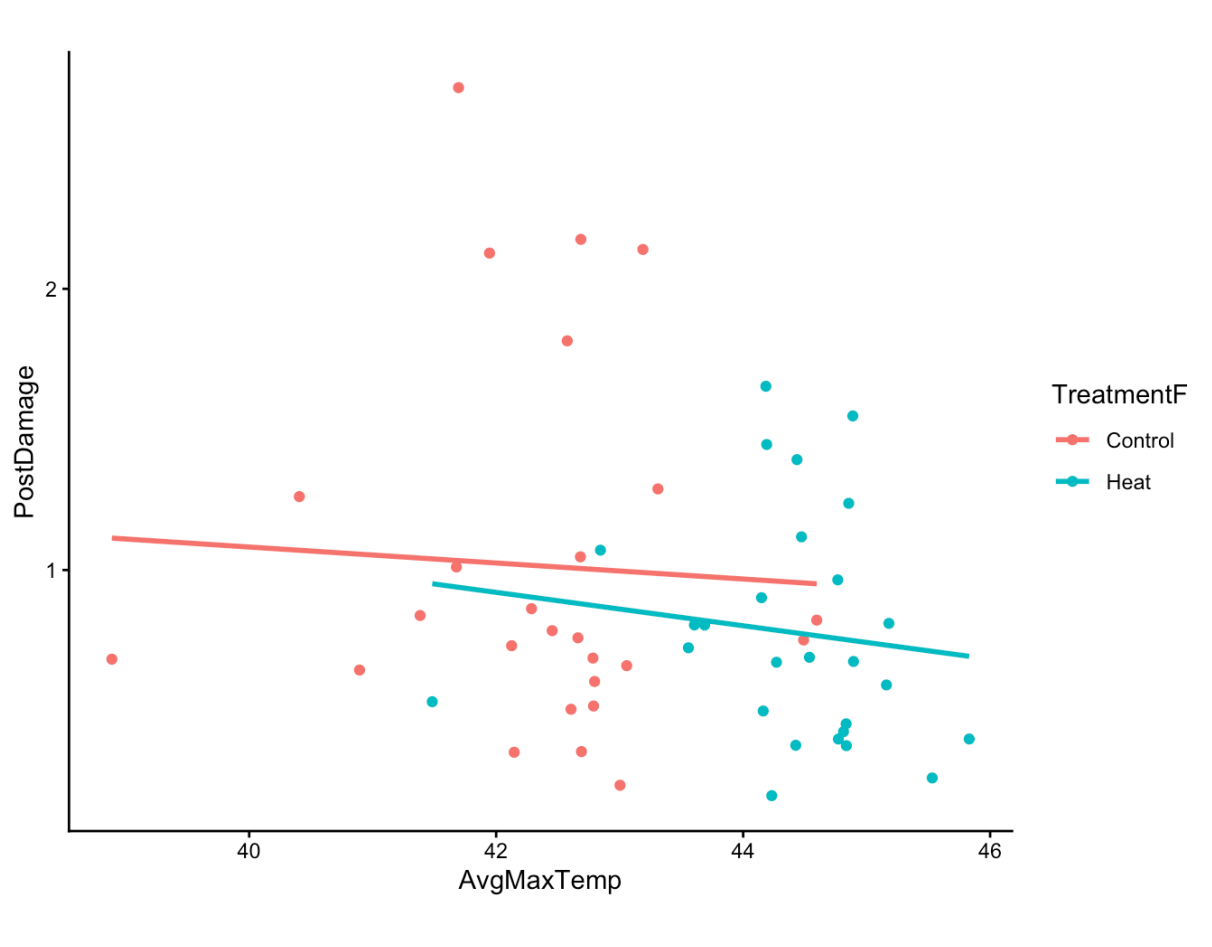
Initial Data Viz

```
# 1. Damage across timepoints and relationship to treatment and sex
ggplot(datumlong2, aes(x = TimepointF, y = Damage, color = SexF, shape = TreatmentF)) +
  geom_point(position = position_jitterdodge(
      jitter.width = 0.2,
      dodge.width = 0.5 )) +
  theme_classic()
```



```
# 2. Correlation between the average top 10% of body temperature values and post damage
ggplot(datum2, aes(x = AvgMaxTemp, y = PostDamage, color = TreatmentF)) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE) +
  theme_classic()
```

```
`geom_smooth()` using formula = 'y ~ x'
```



Summary Stats

```
# 1. Averages for all variables of interest
table = datumlong2 %>%
  group_by(TreatmentF, TimepointF, SexF) %>%
  summarise(
    meanDamage = mean(Damage, na.rm = TRUE),
    sdDamage = sd(Damage, na.rm = TRUE),
    n = n(),
    .groups = "drop"
  )

print(table)
```

```
  TreatmentF TimepointF SexF meanDamage  sdDamage  n
1    Control       Post    F  0.9471522 0.6194113 18
2    Control       Post    M  0.9079955 0.5844157 18
3    Control        Pre    F  0.7531928 0.4109044 20
4    Control        Pre    M  0.7283742 0.2429014 14
5       Heat       Post    F  0.7858846 0.3918378 19
6       Heat       Post    M  0.7021685 0.3720040 16
7       Heat        Pre    F  0.7739065 0.5045865 22
8       Heat        Pre    M  0.7318462 0.3025085 20
```

Analysis - Damage across time points

```
# 1. Running the models

  #Start with global linear mixed effects model to assess damage levels
results = lmer(Damage ~ TreatmentF * SexF * TimepointF + (1|ID),
               data = datumlong2, na.action = na.omit, REML = FALSE)
summary(results)
```

```
Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's
  method [lmerModLmerTest]
Formula: Damage ~ TreatmentF * SexF * TimepointF + (1 | ID)
   Data: datumlong2

      AIC       BIC    logLik -2*log(L)  df.resid
    178.6     208.5     -79.3     158.6       137

Scaled residuals:
    Min      1Q  Median      3Q     Max
-2.8058 -0.5084 -0.1013  0.3446  3.3712

Random effects:
 Groups   Name        Variance Std.Dev.
 ID       (Intercept) 0.10015  0.3165
 Residual             0.09619  0.3101
Number of obs: 147, groups:  ID, 83

Fixed effects:
                                   Estimate Std. Error        df t value
(Intercept)                         0.94792    0.10307 125.24711   9.197
TreatmentFHeat                     -0.18113    0.14316 128.08495  -1.265
SexFM                              -0.03343    0.14623 123.31475  -0.229
TimepointFPre                      -0.19473    0.10211  62.15340  -1.907
TreatmentFHeat:SexFM               -0.06139    0.20682 128.66283  -0.297
TreatmentFHeat:TimepointFPre        0.22241    0.14308  64.04344   1.554
SexFM:TimepointFPre                 0.06848    0.15452  66.38621   0.443
TreatmentFHeat:SexFM:TimepointFPre  -0.03357    0.21381  66.88307  -0.157
                                  Pr(>|t|)
(Intercept)                       1.03e-15 ***
TreatmentFHeat                      0.2081
SexFM                               0.8195
TimepointFPre                       0.0611 .
TreatmentFHeat:SexFM                0.7671
TreatmentFHeat:TimepointFPre        0.1250
SexFM:TimepointFPre                 0.6591
TreatmentFHeat:SexFM:TimepointFPre   0.8757
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
            (Intr) TrtmFH SexFM  TmpnFP TrFH:SFM TFH:TF SFM:TF
TreatmntFHt -0.720
SexFM       -0.705  0.507
TimepontFPr -0.534  0.384  0.376
TrtmnFH:SFM  0.498 -0.692 -0.707 -0.266
TrtmnFH:TFP  0.381 -0.545 -0.268 -0.714  0.377
SxFM:TmpnFP  0.353 -0.254 -0.495 -0.661  0.350    0.472
TFH:SFM:TFP -0.255  0.365  0.358  0.478 -0.532   -0.669 -0.723
```

```
Anova(results, type = "III")
```

```
Analysis of Deviance Table (Type III Wald chisquare tests)

Response: Damage
                            Chisq Df Pr(>Chisq)
(Intercept)               84.5776  1    < 2e-16 ***
TreatmentF                 1.6010  1    0.20576
SexF                       0.0523  1    0.81917
TimepointF                 3.6371  1    0.05651 .
TreatmentF:SexF            0.0881  1    0.76660
TreatmentF:TimepointF      2.4163  1    0.12008
SexF:TimepointF            0.1964  1    0.65765
TreatmentF:SexF:TimepointF  0.0247  1    0.87524
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# 2. The three way interaction was not significant, so it was removed and all two way in
results2 = lmer(Damage ~ TreatmentF * SexF + TreatmentF * TimepointF +
                  SexF * TimepointF + (1|ID), data = datumlong2, na.action = na.omit,
summary(results2)
```

```
Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's
  method [lmerModLmerTest]
Formula: Damage ~ TreatmentF * SexF + TreatmentF * TimepointF + SexF *
    TimepointF + (1 | ID)
   Data: datumlong2

      AIC       BIC    logLik -2*log(L)  df.resid
    176.6     203.5     -79.3     158.6       138

Scaled residuals:
    Min      1Q  Median      3Q     Max
-2.8147 -0.5202 -0.0854  0.3515  3.3818

Random effects:
 Groups   Name        Variance Std.Dev.
 ID       (Intercept) 0.09991  0.3161
 Residual             0.09634  0.3104
Number of obs: 147, groups:  ID, 83

Fixed effects:
                             Estimate Std. Error        df t value Pr(>|t|)
(Intercept)                   0.94381    0.09965 115.22962   9.472 4.34e-16
TreatmentFHeat               -0.17292    0.13326 107.10396  -1.298   0.1972
SexFM                        -0.02526    0.13652 101.18775  -0.185   0.8536
TimepointFPre                -0.18710    0.08978  63.65926  -2.084   0.0412
TreatmentFHeat:SexFM         -0.07858    0.17500  77.44862  -0.449   0.6547
TreatmentFHeat:TimepointFPre  0.20735    0.10640  66.49030   1.949   0.0555
SexFM:TimepointFPre           0.05090    0.10688  67.08481   0.476   0.6354

(Intercept)                  ***
TreatmentFHeat
SexFM
TimepointFPre                *
TreatmentFHeat:SexFM
TreatmentFHeat:TimepointFPre .
SexFM:TimepointFPre
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
            (Intr) TrtmFH SexFM  TmpnFP TFH:SF TFH:TF
TreatmntFHt -0.696
SexFM       -0.680  0.433
TimepontFPr -0.485  0.257  0.250
TrtmnFH:SFM  0.443 -0.632 -0.653 -0.016
TrtmnFH:TFP  0.293 -0.435 -0.042 -0.604  0.033
SxFM:TmpnFP  0.252  0.015 -0.367 -0.520 -0.059 -0.024
```

```
Anova(results2, type = "III")
```

```
Analysis of Deviance Table (Type III Wald chisquare tests)

Response: Damage
                       Chisq Df Pr(>Chisq)
(Intercept)          89.7121  1    < 2e-16 ***
TreatmentF            1.6839  1    0.19440
SexF                  0.0342  1    0.85322
TimepointF            4.3429  1    0.03716 *
TreatmentF:SexF       0.2016  1    0.65342
TreatmentF:TimepointF  3.7977  1    0.05132 .
SexF:TimepointF       0.2268  1    0.63389
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# 3. The Sex*Timepoint interaction was not significant, so it was removed from the model
results3 = lmer(Damage ~ TreatmentF * SexF + TreatmentF * TimepointF +
                  (1|ID), data = datumlong2, na.action = na.omit, REML = FALSE)
summary(results3)
```

```
Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's
  method [lmerModLmerTest]
Formula: Damage ~ TreatmentF * SexF + TreatmentF * TimepointF + (1 | ID)
   Data: datumlong2

      AIC       BIC    logLik -2*log(L)  df.resid
    174.8     198.8     -79.4     158.8       139

Scaled residuals:
    Min      1Q  Median      3Q     Max
-2.8381 -0.5057 -0.1172  0.3410  3.4102

Random effects:
 Groups   Name        Variance Std.Dev.
 ID       (Intercept) 0.09923  0.3150
 Residual             0.09695  0.3114
Number of obs: 147, groups:  ID, 83

Fixed effects:
                              Estimate Std. Error         df t value Pr(>|t|)
(Intercept)                   0.931912   0.096377 104.630425   9.669 3.49e-16
TreatmentFHeat               -0.173831   0.133188 107.347025  -1.305   0.1946
SexFM                        -0.001603   0.126868  76.809103  -0.013   0.9900
TimepointFPre                -0.165012   0.076922  65.689895  -2.145   0.0356
TreatmentFHeat:SexFM         -0.073358   0.174508  77.214262  -0.420   0.6754
TreatmentFHeat:TimepointFPre  0.208436   0.106679  66.563421   1.954   0.0549

(Intercept)                  ***
TreatmentFHeat
SexFM
TimepointFPre                *
TreatmentFHeat:SexFM
TreatmentFHeat:TimepointFPre .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
            (Intr) TrtmFH SexFM  TmpnFP TFH:SF
TreatmntFHt -0.724
SexFM       -0.652  0.472
TimepontFPr -0.430  0.311  0.076
TrtmnFH:SFM  0.474 -0.632 -0.727 -0.055
TrtmnFH:TFP  0.310 -0.437 -0.054 -0.721  0.032
```

```
Anova(results3, type = "III")
```

```
Analysis of Deviance Table (Type III Wald chisquare tests)

Response: Damage
                       Chisq Df Pr(>Chisq)
(Intercept)          93.4975  1    < 2e-16 ***
TreatmentF            1.7034  1    0.19184
SexF                  0.0002  1    0.98992
TimepointF            4.6019  1    0.03194 *
TreatmentF:SexF       0.1767  1    0.67422
TreatmentF:TimepointF  3.8176  1    0.05072 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# 4. The Treatment*Sex interaction was not significant, so it was removed from the model
results4 = lmer(Damage ~ TreatmentF * TimepointF + SexF + (1|ID), data = datumlong2,
                na.action = na.omit, REML = FALSE)
summary(results4)
```

```
Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's
  method [lmerModLmerTest]
Formula: Damage ~ TreatmentF * TimepointF + SexF + (1 | ID)
   Data: datumlong2

      AIC       BIC    logLik -2*log(L)  df.resid
    173.0     193.9     -79.5     159.0       140

Scaled residuals:
    Min      1Q  Median      3Q     Max
-2.8500 -0.5140 -0.1114  0.3308  3.3843

Random effects:
 Groups   Name        Variance Std.Dev.
 ID       (Intercept) 0.0991   0.3148
 Residual             0.0972   0.3118
Number of obs: 147, groups:  ID, 83

Fixed effects:
                             Estimate Std. Error        df t value Pr(>|t|)
(Intercept)                   0.95111    0.08490 112.29575  11.203   <2e-16
TreatmentFHeat               -0.20914    0.10330 128.95555  -2.025   0.0450
TimepointFPre                -0.16683    0.07690  65.71642  -2.169   0.0337
SexFM                        -0.04039    0.08712  77.03983  -0.464   0.6442
TreatmentFHeat:TimepointFPre  0.20985    0.10676  66.43885   1.966   0.0535

(Intercept)                  ***
TreatmentFHeat               *
TimepointFPre                *
SexFM
TreatmentFHeat:TimepointFPre .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
            (Intr) TrtmFH TmpnFP SexFM
TreatmntFHt -0.622
TimepontFPr -0.460  0.357
SexFM       -0.508  0.023  0.052
TrtmnFH:TFP  0.335 -0.538 -0.721 -0.045
```

```
Anova(results4, type = "III")
```

```
Analysis of Deviance Table (Type III Wald chisquare tests)

Response: Damage
                        Chisq Df Pr(>Chisq)
(Intercept)           125.5155  1    < 2e-16 ***
TreatmentF              4.0987  1    0.04292 *
TimepointF              4.7063  1    0.03005 *
SexF                    0.2149  1    0.64293
TreatmentF:TimepointF   3.8639  1    0.04934 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# 5. The Treatment*Sex interaction was p=0.052 , so to ensure that is not significant, w
results5 = lmer(Damage ~ TreatmentF + TimepointF + SexF + (1|ID), data = datumlong2,
                na.action = na.omit, REML = FALSE)
summary(results5)
```

```
Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's
  method [lmerModLmerTest]
Formula: Damage ~ TreatmentF + TimepointF + SexF + (1 | ID)
   Data: datumlong2

      AIC       BIC    logLik -2*log(L)  df.resid
    174.7     192.7     -81.4     162.7       141

Scaled residuals:
    Min      1Q  Median      3Q     Max
-2.8995 -0.4906 -0.1338  0.3475  3.4626

Random effects:
 Groups   Name        Variance Std.Dev.
 ID       (Intercept) 0.09411  0.3068
 Residual             0.10391  0.3224
Number of obs: 147, groups:  ID, 83

Fixed effects:
               Estimate Std. Error       df t value Pr(>|t|)
(Intercept)     0.89548    0.07996 93.85580  11.199   <2e-16 ***
TreatmentFHeat -0.09890    0.08678 76.65616  -1.140    0.258
TimepointFPre  -0.05962    0.05501 67.16758  -1.084    0.282
SexFM          -0.03316    0.08671 76.87230  -0.382    0.703
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
            (Intr) TrtmFH TmpnFP
TreatmntFHt -0.552
TimepontFPr -0.345 -0.052
SexFM       -0.521 -0.002  0.028
```

```
Anova(results5, type = "III")
```

```
Analysis of Deviance Table (Type III Wald chisquare tests)

Response: Damage
               Chisq Df Pr(>Chisq)
(Intercept) 125.4228  1     <2e-16 ***
TreatmentF    1.2988  1     0.2544
TimepointF    1.1748  1     0.2784
SexF          0.1463  1     0.7021
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
#Use AIC to test which one of these models has the best fit
AIC(results, results2, results3, results4, results5)
```

```
         df      AIC
results  10 178.5777
results2  9 176.6022
results3  8 174.8278
results4  7 173.0043
results5  6 174.7314
```

```
#According to AIC scores, use 'results4' for testing the assumptions

results_final = lmer(Damage ~ TreatmentF * TimepointF + SexF + (1|ID), data = datumlong2
```

```r
                na.action = na.omit, REML = TRUE)
summary(results_final)
```

```
Linear mixed model fit by REML. t-tests use Satterthwaite's method [
lmerModLmerTest]
Formula: Damage ~ TreatmentF * TimepointF + SexF + (1 | ID)
   Data: datumlong2

REML criterion at convergence: 176.1

Scaled residuals:
    Min      1Q  Median      3Q     Max 
-2.8123 -0.5048 -0.1089  0.3272  3.3331 

Random effects:
 Groups   Name        Variance Std.Dev.
 ID       (Intercept) 0.1036   0.3219  
 Residual             0.1000   0.3163  
Number of obs: 147, groups:  ID, 83

Fixed effects:
                              Estimate Std. Error        df t value Pr(>|t|)
(Intercept)                    0.95110    0.08649 107.92394  10.996   <2e-16
TreatmentFHeat                -0.20936    0.10520 123.98493  -1.990   0.0488
TimepointFPre                 -0.16665    0.07804  63.93617  -2.136   0.0366
SexFM                         -0.04034    0.08884  74.28932  -0.454   0.6511
TreatmentFHeat:TimepointFPre   0.20997    0.10834  64.63317   1.938   0.0570
                                
(Intercept)                  ***
TreatmentFHeat               *  
TimepointFPre                *  
SexFM                           
TreatmentFHeat:TimepointFPre .  
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
            (Intr) TrtmFH TmpnFP SexFM 
TreatmntFHt -0.621                     
TimepontFPr -0.458  0.356              
SexFM       -0.508  0.023  0.052       
TrtmnFH:TFP  0.334 -0.536 -0.721 -0.045
```

```r
Anova(results_final, type = "III")
```

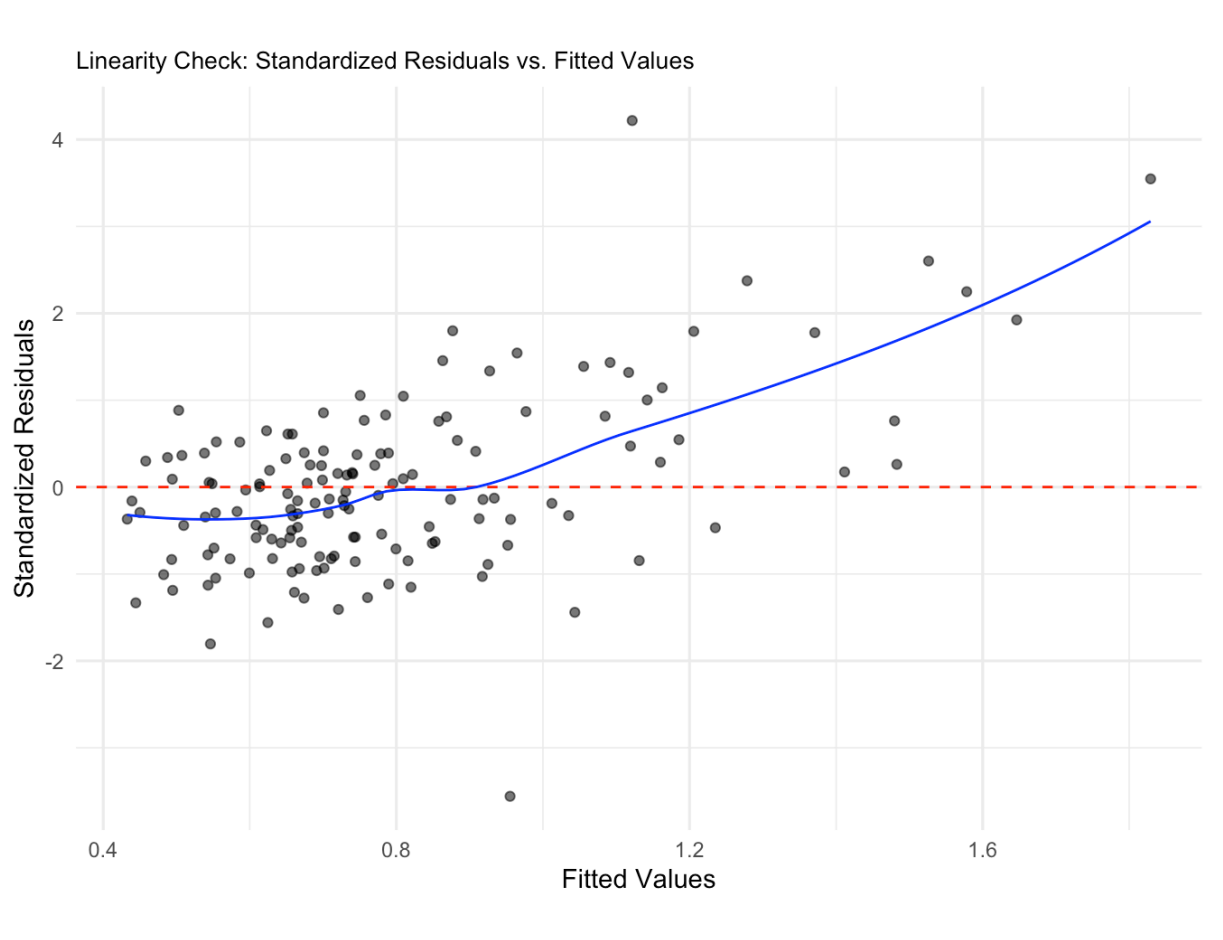
```
Analysis of Deviance Table (Type III Wald chisquare tests)

Response: Damage
                         Chisq Df Pr(>Chisq)    
(Intercept)           120.9227  1    < 2e-16 ***
TreatmentF              3.9605  1    0.04658 *  
TimepointF              4.5605  1    0.03272 *  
SexF                    0.2062  1    0.64979    
TreatmentF:TimepointF   3.7565  1    0.05260 .  
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
#Assumptions of the model


# 1. Check the VIF of this model
vif(results_final)
```

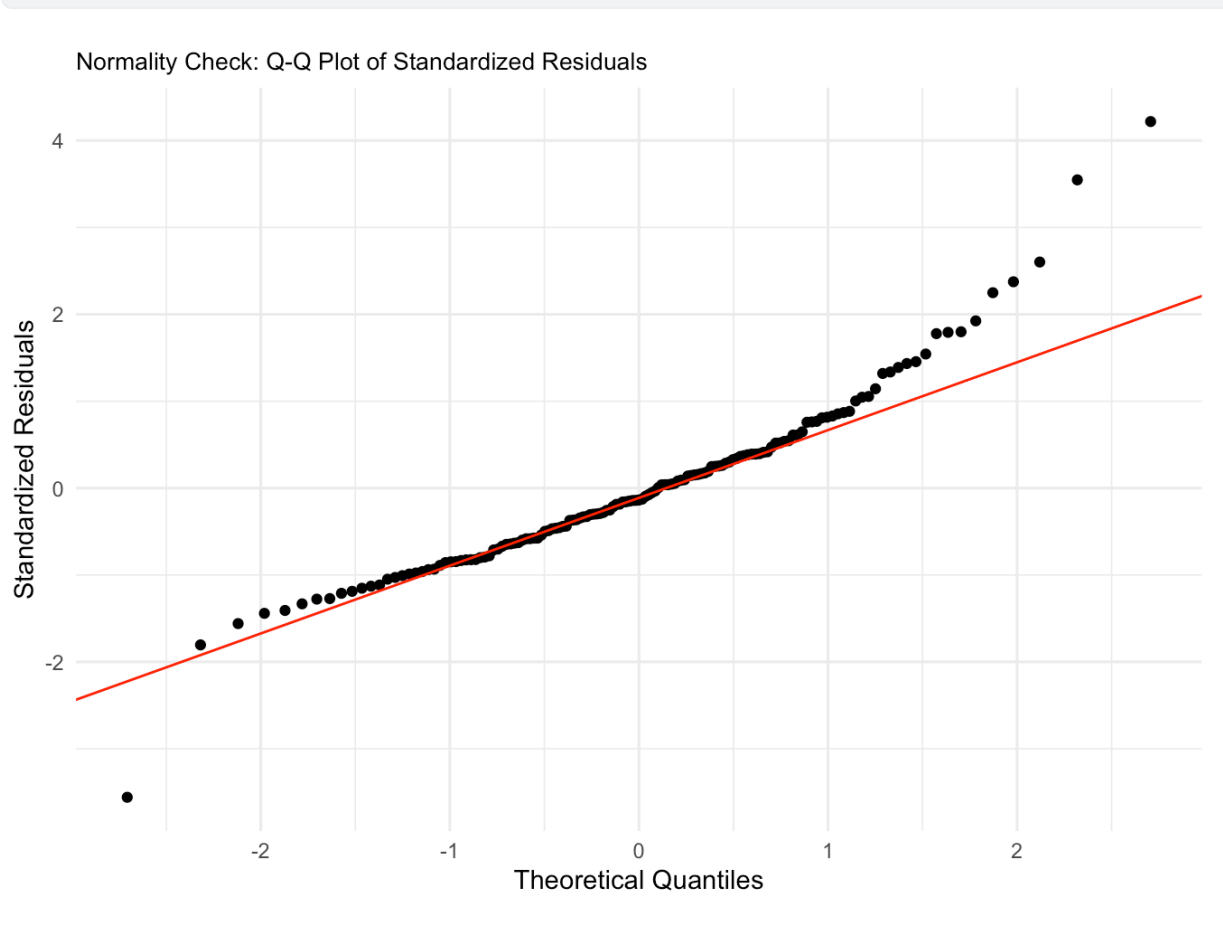
```
           TreatmentF            TimepointF                  SexF 
             1.406170              2.088099              1.002821 
TreatmentF:TimepointF 
             2.555517 
```

```r
  #VIF = 1: No multicollinearity
  #VIF between 1 and 5: Generally considered acceptable or moderate multicollinearity
  #VIF > 5 (or > 10): Indicates potentially high multicollinearity

# 2. Check other assumptions of a linear model

# --- Extract Residuals and Fitted Values ---
#1. Extract the data frame used by the model
datum_clean = model.frame(results_final)

# 2. Calculate and assign the fitted and residual values to this clean data frame
datum_clean$fitted <- fitted(results_final)
datum_clean$raw_resid <- residuals(results_final)
datum_clean$std_resid <- datum_clean$raw_resid / sd(datum_clean$raw_resid, na.rm = TRUE)

# 3. Use the studentized residuals method from the 'lme4' utility functions
# Although 'rstandard' doesn't work, 'residuals' is robust.
# If you really need a standardized residual, use this function:
datum_clean$std_resid <- datum_clean$raw_resid / sd(datum_clean$raw_resid, na.rm = TRUE)


# --- 2. Create the Plots ---

# A. Linearity and Homoscedasticity Plot (Residuals vs. Fitted)
ggplot(datum_clean, aes(x = fitted, y = std_resid)) +
  geom_point(alpha = 0.6) +
  # Add the ideal reference line at y=0
  geom_hline(yintercept = 0, linetype = "dashed", color = "red") +
  # Add a LOESS smoothing line (red line in performance)
  geom_smooth(method = "loess", se = FALSE, color = "blue", linewidth = 0.5) +
  labs(
    title = "Linearity Check: Standardized Residuals vs. Fitted Values",
    x = "Fitted Values",
    y = "Standardized Residuals"
  ) +
  theme_minimal() +
  theme(plot.title = element_text(size = 10))
```

```
`geom_smooth()` using formula = 'y ~ x'
```

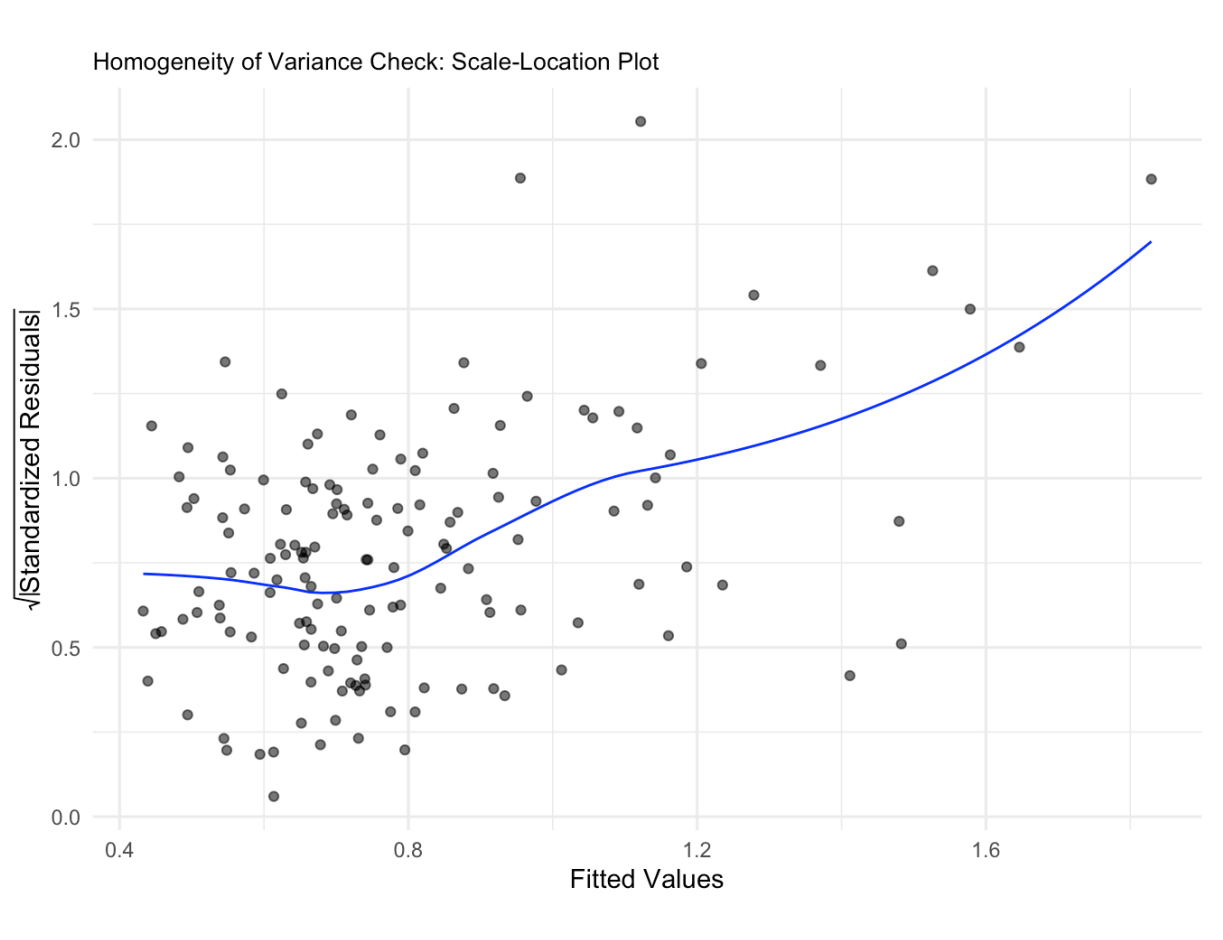


```r
# B. Normality Check (Quantile-Quantile Plot)
ggplot(datum_clean, aes(sample = std_resid)) +
  # Create the Q-Q plot
  stat_qq() +
  # Add the reference line (the ideal normal distribution line)
  stat_qq_line(color = "red") +
  labs(
    title = "Normality Check: Q-Q Plot of Standardized Residuals",
    x = "Theoretical Quantiles",
    y = "Standardized Residuals"
  ) +
  theme_minimal() +
  theme(plot.title = element_text(size = 10))
```

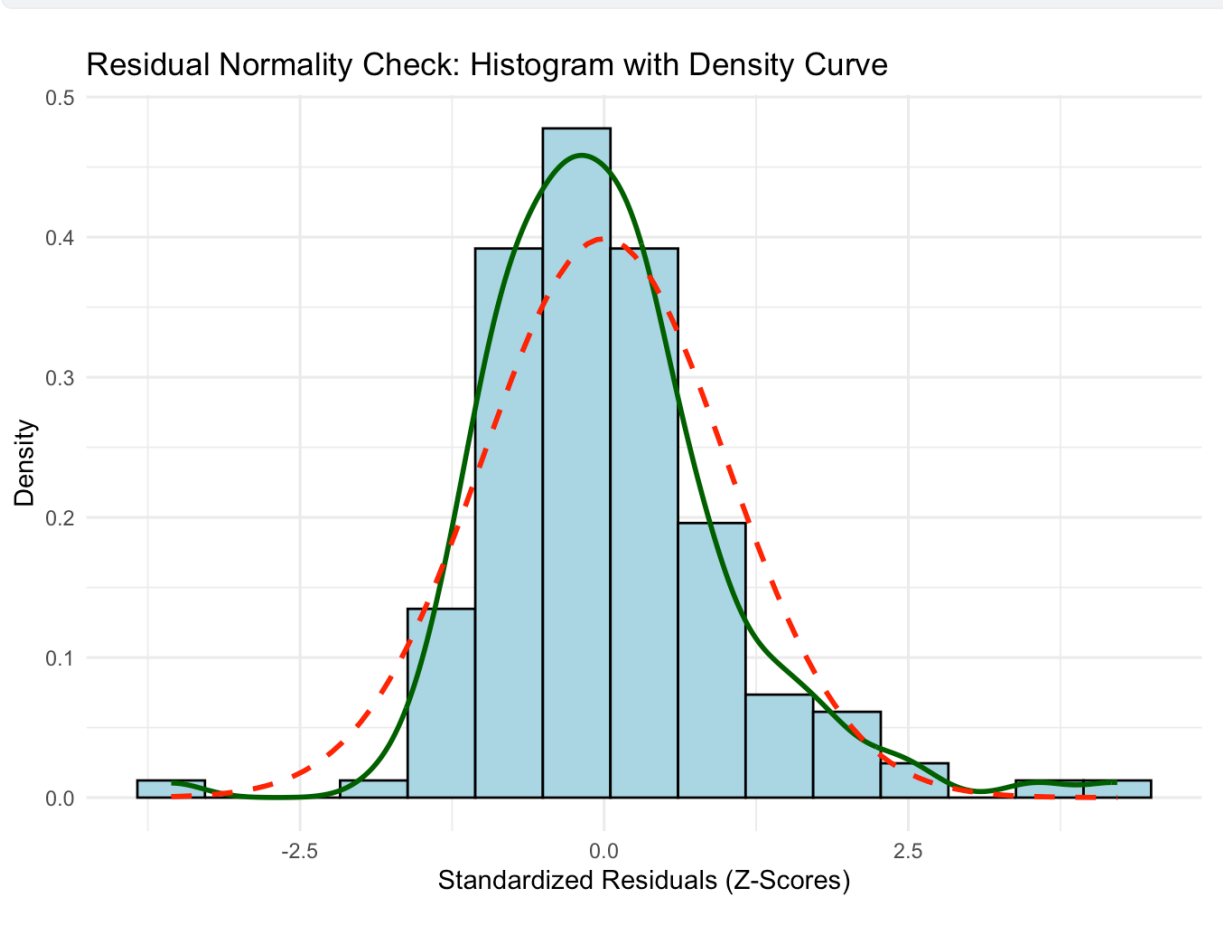


```r
# C. Scale-Location Plot (Homogeneity of Variance Check)
# This uses the square root of the absolute standardized residuals
ggplot(datum_clean, aes(x = fitted, y = sqrt(abs(std_resid)))) +
  geom_point(alpha = 0.6) +
  # Add the ideal flat reference line
  geom_smooth(method = "loess", se = FALSE, color = "blue", linewidth = 0.5) +
  labs(
    title = "Homogeneity of Variance Check: Scale-Location Plot",
    x = "Fitted Values",
    y = expression(sqrt(abs("Standardized Residuals")))
  ) +
  theme_minimal() +
  theme(plot.title = element_text(size = 10))
```

```
`geom_smooth()` using formula = 'y ~ x'
```



```r
#D. Histogram of residuals
# Calculate mean and standard deviation of residuals for the density curve
resid_mean <- mean(datum_clean$std_resid, na.rm = TRUE)
resid_sd <- sd(datum_clean$std_resid, na.rm = TRUE)

ggplot(datum_clean, aes(x = std_resid)) +
  # 1. Create the Histogram (using density scale)
  geom_histogram(aes(y = after_stat(density)),
                 bins = 15, # Adjust bin count for best visualization
                 color = "black",
                 fill = "lightblue") +

  # 2. Overlay the Density of the Residuals (Green Line)
  geom_density(color = "darkgreen", linewidth = 1) +

  # 3. Overlay the Theoretical Normal Density Curve (Red Dashed Line)
  stat_function(fun = dnorm,
                args = list(mean = resid_mean, sd = resid_sd),
                color = "red",
                linewidth = 1,
                linetype = "dashed") +

  labs(
    title = "Residual Normality Check: Histogram with Density Curve",
    x = "Standardized Residuals (Z-Scores)",
    y = "Density"
  ) +
  theme_minimal()
```



Conclusion: Histogram looks good. Overall, this model is a good fit.

Next Steps: Continue to post-hoc analysis.

```r
emms_interaction = emmeans(results_final, ~ TreatmentF * TimepointF)
interaction_pairs = pairs(emms_interaction, adjust = "sidak")
print(interaction_pairs)
```

```
 contrast                    estimate     SE    df t.ratio p.value
 Control Post - Heat Post    0.209358 0.1050 126.1   1.988  0.2604
 Control Post - Control Pre  0.166653 0.0782  69.0   2.131  0.2006
 Control Post - Heat Pre     0.166038 0.1020 120.6   1.632  0.4872
 Heat Post - Control Pre    -0.042706 0.1070 127.8  -0.401  0.9991
 Heat Post - Heat Pre       -0.043321 0.0753  70.5  -0.575  0.9934
 Control Pre - Heat Pre     -0.000615 0.1030 122.8  -0.006  1.0000

Results are averaged over the levels of: SexF 
Degrees-of-freedom method: kenward-roger 
P value adjustment: sidak method for 6 tests 
```

```r
cld_output = cld(emms_interaction,
                 adjust = "sidak",
                 Letters = letters)

print(cld_output)
```

```
  TreatmentF TimepointF    emmean         SE       df  lower.CL  upper.CL
2       Heat       Post 0.7215771 0.07445444 130.2289 0.5335184 0.9096357
3    Control        Pre 0.7642828 0.07633590 125.6381 0.5713732 0.9571924
4       Heat        Pre 0.7648977 0.06929134 119.3218 0.5896558 0.9401395
1    Control       Post 0.9309354 0.07452638 121.5577 0.7425068 1.1193640
  .group
2      a
3      a
4      a
1      a
```

Analysis - Damage across time % change

```r
# 1. Making the column

datum2$DeltaDamage = datum2$PostDamage/datum2$PreDamage

# 2. Running the models

  #Start with global linear model to assess damage levels
results = lm(DeltaDamage ~ TreatmentF * SexF, data = datum2, na.action = na.omit)
summary(results)
```

```

Call:
lm(formula = DeltaDamage ~ TreatmentF * SexF, data = datum2, 
    na.action = na.omit)

Residuals:
    Min      1Q  Median      3Q     Max 
-3.2225 -1.0370 -0.4167 -0.0569 29.5219 

Coefficients:
                    Estimate Std. Error t value Pr(>|t|)   
(Intercept)            3.663      1.306   2.805  0.00741 **
TreatmentFHeat        -2.193      1.814  -1.209  0.23300   
SexFM                 -2.412      1.981  -1.218  0.22966   
TreatmentFHeat:SexFM   1.833      2.712   0.676  0.50265   
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.709 on 45 degrees of freedom
  (4 observations deleted due to missingness)
Multiple R-squared:  0.05615,	Adjusted R-squared:  -0.006773 
F-statistic: 0.8924 on 3 and 45 DF,  p-value: 0.4524
```

```r
Anova(results, type = "III")
```

```
Anova Table (Type III tests)

Response: DeltaDamage
                Sum Sq Df F value   Pr(>F)   
(Intercept)     174.44  1  7.8669 0.007408 **
TreatmentF       32.41  1  1.4615 0.233001   
SexF             32.88  1  1.4829 0.229663   
TreatmentF:SexF  10.13  1  0.4566 0.502652   
Residuals       997.81 45                    
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
  #The interaction was not significant, so it was removed from the model
results2 = lm(DeltaDamage ~ TreatmentF + SexF, data = datum2, na.action = na.omit)
summary(results2)
```

```

Call:
lm(formula = DeltaDamage ~ TreatmentF + SexF, data = datum2, 
    na.action = na.omit)

Residuals:
    Min      1Q  Median      3Q     Max 
-2.7975 -1.3150 -0.8114  0.3380 29.9469 

Coefficients:
               Estimate Std. Error t value Pr(>|t|)   
(Intercept)       3.238      1.138   2.846  0.00659 **
TreatmentFHeat   -1.373      1.340  -1.024  0.31106   
SexFM            -1.434      1.345  -1.067  0.29174   
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.681 on 46 degrees of freedom
  (4 observations deleted due to missingness)
Multiple R-squared:  0.04657,	Adjusted R-squared:  0.005119 
F-statistic: 1.123 on 2 and 46 DF,  p-value: 0.3339
```

```r
Anova(results2, type = "III")
```

```
Anova Table (Type III tests)

Response: DeltaDamage
             Sum Sq Df F value   Pr(>F)   
(Intercept)  177.47  1  8.0991 0.006593 **
TreatmentF    22.99  1  1.0492 0.311058   
SexF          24.93  1  1.1375 0.291741   
Residuals   1007.94 46                    
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
  #Neither Sex nor Treatment affected the outcome, so each main effect was ran by itself
results3 = lm(DeltaDamage ~ SexF, data = datum2, na.action = na.omit)
summary(results3)
```

```

Call:
lm(formula = DeltaDamage ~ SexF, data = datum2, na.action = na.omit)

Residuals:
    Min      1Q  Median      3Q     Max 
-2.0927 -1.4921 -0.7438 -0.1493 30.6588 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)   
(Intercept)   2.5262     0.9013   2.803  0.00734 **
SexFM        -1.4714     1.3451  -1.094  0.27959   
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.683 on 47 degrees of freedom
  (4 observations deleted due to missingness)
Multiple R-squared:  0.02483,	Adjusted R-squared:  0.004078 
F-statistic: 1.197 on 1 and 47 DF,  p-value: 0.2796
```

```r
Anova(results3, type = "III")
```

```
Anova Table (Type III tests)

Response: DeltaDamage
             Sum Sq Df F value   Pr(>F)   
(Intercept)  172.30  1  7.8553 0.007339 **
SexF          26.25  1  1.1966 0.279587   
Residuals   1030.93 47                    
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
  #Neither Sex nor Treatment affected the outcome, so each main effect was ran by itself
results4 = lm(DeltaDamage ~ TreatmentF, data = datum2, na.action = na.omit)
summary(results4)
```

```

Call:
lm(formula = DeltaDamage ~ TreatmentF, data = datum2, na.action = na.omit)

Residuals:
    Min      1Q  Median      3Q     Max 
-2.1738 -1.5789 -0.5621 -0.1494 30.5705 

Coefficients:
               Estimate Std. Error t value Pr(>|t|)  
(Intercept)      2.6144     0.9775   2.675   0.0103 *
TreatmentFHeat  -1.4114     1.3419  -1.052   0.2983  
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.688 on 47 degrees of freedom
  (4 observations deleted due to missingness)
Multiple R-squared:  0.02299,	Adjusted R-squared:  0.002207 
F-statistic: 1.106 on 1 and 47 DF,  p-value: 0.2983
```

```r
Anova(results4, type = "III")
```

```
Anova Table (Type III tests)

Response: DeltaDamage
             Sum Sq Df F value  Pr(>F)  
(Intercept)  157.21  1  7.1538 0.01026 *
TreatmentF    24.31  1  1.1062 0.29829  
Residuals   1032.86 47                  
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The outcomes here are not any different than when we include time point, and so I would like to keep time point in the model.

Analysis - Body Temperature and Damage Correlation

```r
# 1. Run Pearson's Correlation Test for both heated and control groups
pearson_result = cor.test(
  x = datum2$PostDamage,
  y = datum2$AvgMaxTemp,
  method = "pearson"
)
  # View the results
print(pearson_result)
```

```

Results of Hypothesis Test
--------------------------

Null Hypothesis:                 correlation = 0

Alternative Hypothesis:          True correlation is not equal to 0

Test Name:                       Pearson's product-moment correlation

Estimated Parameter(s):          cor = -0.2059416

Data:                            datum2$PostDamage and datum2$AvgMaxTemp

Test Statistic:                  t = -1.502933

Test Statistic Parameter:        df = 51

P-value:                         0.1390236

95% Confidence Interval:         LCL = -0.45112369
                                 UCL =  0.06814556
```

```r
# 2.  Run Pearson's Correlation Test heated group
datumHeated = datum2 %>%
 filter(TreatmentF == "Heat")

pearson_resultHeat = cor.test(
  x = datumHeated$PostDamage,
  y = datumHeated$AvgMaxTemp,
  method = "pearson"
)
  # View the results
print(pearson_resultHeat)
```

```

Results of Hypothesis Test
--------------------------

Null Hypothesis:                 correlation = 0

Alternative Hypothesis:          True correlation is not equal to 0

Test Name:                       Pearson's product-moment correlation

Estimated Parameter(s):          cor = -0.125062

Data:                            datumHeated$PostDamage and datumHeated$AvgMaxTemp

Test Statistic:                  t = -0.6302581

Test Statistic Parameter:        df = 25

P-value:                         0.5342448

95% Confidence Interval:         LCL = -0.4821611
                                 UCL =  0.2676732
```

```r
# 2.  Run Pearson's Correlation Test control group
datumControl = datum2 %>%
 filter(TreatmentF == "Control")

pearson_resultControl = cor.test(
  x = datumControl$PostDamage,
  y = datumControl$AvgMaxTemp,
  method = "pearson"
)
  # View the results
print(pearson_resultControl)
```

```

Results of Hypothesis Test
--------------------------

Null Hypothesis:                 correlation = 0

Alternative Hypothesis:          True correlation is not equal to 0

Test Name:                       Pearson's product-moment correlation

Estimated Parameter(s):          cor = -0.05057064

Data:                            datumControl$PostDamage and datumControl$AvgMaxTemp

Test Statistic:                  t = -0.2480619

Test Statistic Parameter:        df = 24

P-value:                         0.8061966

95% Confidence Interval:         LCL = -0.4295091
                                 UCL =  0.3435102
```

Nothing was significant here, which makes sense based on previous models

Manuscript Graph

```r
# 1. Add the means and 95% CIs to the dataframe for the variables we are interested in

datumlong2_with_summary = datumlong2 %>%

  # 1. Group the data by the interaction terms
  group_by(TreatmentF, TimepointF) %>%

  # 2. Calculate the group-level statistics (Mean, CI, etc.)
  mutate(
    N_group = n(),
    Mean_Damage_Group = mean(Damage, na.rm = TRUE),
    SD_Damage_Group = sd(Damage, na.rm = TRUE),
    SE_Damage_Group = SD_Damage_Group / sqrt(N_group),
    CI_Margin_Damage = 1.96 * SE_Damage_Group,
    CI_Upper_Group = Mean_Damage_Group + CI_Margin_Damage,
    CI_Lower_Group = Mean_Damage_Group - CI_Margin_Damage
  ) %>%
  ungroup()
  #View the dataframe that was created
head(datumlong2_with_summary)
```

```
# A tibble: 6 × 35
     ID Sex   Treatment Incubator Side  Timepoint Died  Date     Mass Glucose
  <int> <chr> <chr>     <chr>     <chr> <chr>     <chr> <chr>   <dbl> <chr>  
1  4819 M     Control   Aphrodite Right Pre       No    3/5/25   15.6 198.5  
2  4820 M     Control   Zeus      Right Pre       No    3/7/25   14.1 191.5  
3  4823 M     Heat      Aphrodite Right Pre       No    3/20/25  14.4 239    
4  4824 M     Heat      Apollo    Left  Pre       No    3/8/25   14.5 226.5  
5  4825 M     Heat      Aphrodite Right Pre       No    3/9/25   13.3 168.5  
6  4826 M     Control   Apollo    Left  Pre       No    3/11/25  14.8 192    
# ℹ 25 more variables: Ketones <chr>, HCT <chr>, CORT <dbl>, Damage <dbl>,
#   X <lgl>, X.1 <lgl>, X.2 <lgl>, X.3 <lgl>, X.4 <lgl>, X.5 <lgl>, X.6 <lgl>,
#   X.7 <lgl>, X.8 <lgl>, X.9 <lgl>, X.10 <lgl>, SexF <fct>, TreatmentF <fct>,
#   TimepointF <fct>, N_group <int>, Mean_Damage_Group <dbl>,
#   SD_Damage_Group <dbl>, SE_Damage_Group <dbl>, CI_Margin_Damage <dbl>,
#   CI_Upper_Group <dbl>, CI_Lower_Group <dbl>
```

```r
# 2. Cleaning up data frame before plotting
  # Define the consistent dodge width and rename the data frame for plotting
DODGE_WIDTH = 0.4
plot_data = datumlong2_with_summary

  #Order the time points so that Pre and Post are graphed in the correct order on the pl
plot_data$TreatmentF = as.factor(plot_data$TreatmentF)
plot_data$TimepointF = factor(plot_data$TimepointF, levels = c("Pre", "Post"))

# 3. Plotting the graph

plot = ggplot(data = plot_data, aes(x = TimepointF, y = Damage, color = TreatmentF)) +
  geom_jitter(
    size = 3.5,
    alpha = 0.5,
    position = position_jitterdodge(jitter.width = 0.2, dodge.width = DODGE_WIDTH)
  ) +

  geom_point(
    aes(y = Mean_Damage_Group, group = TreatmentF),
    size = 7.5,
    stroke = 1,
    color = "black",
    position = position_dodge(width = DODGE_WIDTH)
  ) +

  geom_errorbar(
    aes(ymin = CI_Lower_Group, ymax = CI_Upper_Group, group = TreatmentF),
    width = 0.2,
    linewidth = 1,
    color = "black",
    position = position_dodge(width = DODGE_WIDTH)
  ) +

  geom_line(
    aes(y = Mean_Damage_Group, group = TreatmentF),
    linewidth = 1.5,
    position = position_dodge(width = DODGE_WIDTH)
  ) +

  scale_color_manual(
    values = c(
      "Control" = "#AAB8C2",
      "Heat" = "firebrick4"
    )
  ) +

  labs(
    x = "Time Point",
    y = "DNA Damage (AU)",
    color = "Treatment Group"
  ) +
  theme_classic() +

  theme(
    axis.title = element_text(face = "bold", size = 30),
    axis.text = element_text(size = 25),
    legend.justification = c(0.5, 0.5),
    legend.title = element_text(size = 28),
    legend.text = element_text(size = 25),
    strip.text = element_text(size = 20),
    panel.background = element_rect(fill = "transparent", color = NA),
    plot.background = element_rect(fill = "transparent", color = NA),
    legend.background = element_rect(fill = "transparent", color = NA),
    legend.box.background = element_rect(fill = "transparent", color = NA),
    axis.line = element_line(linewidth = 1.5, colour = "black")
  )

# 4. Adding signifigance and sample size

plot1 = plot +
  annotate(
    "text",
    x = c(.9, 1.1, 1.9, 2.1),
    y = c(2.1 + .3, 2.5 + .3, 3.1 + .3, 2 + .3),
    label = c("a", "a", "a", "a"),
    size = 10,
    color = "black") +

   # Add a new layer for the sample sizes
  annotate(
    "text",
    x = c(.9, 1.1, 1.9, 2.1),
    # Adjust the y-coordinates to be slightly lower
    y = c(2.1, 2.5, 3.1, 2),
    # Add your sample sizes here
    label = c("(34)", "(42)", "(36)", "(35)"),
    size = 7.5, # Make the font size slightly smaller
    color = "black")

# 5. Save the graph
ggsave(
  filename = "DamagePlot.png",
  plot = plot1,
  width = 11.04,
  height = 6.49,
  units = "in",
  dpi = 300,
  bg = "transparent"
)
```

Manuscript Graph - With Sex

```
# 1. Add the means and 95% CIs to the dataframe for the variables we are interested in

datumlong2_with_summary = datumlong2 %>%

  # 1. Group the data by the interaction terms
  group_by(TreatmentF, TimepointF) %>%

  # 2. Calculate the group-level statistics (Mean, CI, etc.)
  mutate(
    N_group = n(),
    Mean_Damage_Group = mean(Damage, na.rm = TRUE),
    SD_Damage_Group = sd(Damage, na.rm = TRUE),
    SE_Damage_Group = SD_Damage_Group / sqrt(N_group),
    CI_Margin_Damage = 1.96 * SE_Damage_Group,
    CI_Upper_Group = Mean_Damage_Group + CI_Margin_Damage,
    CI_Lower_Group = Mean_Damage_Group - CI_Margin_Damage
  ) %>%
  ungroup()
  #View the dataframe that was created
head(datumlong2_with_summary)
```

```
# A tibble: 6 × 35
     ID Sex   Treatment Incubator Side  Timepoint Died  Date     Mass Glucose
  <int> <chr> <chr>     <chr>     <chr> <chr>     <chr> <chr>   <dbl> <chr>
1  4819 M     Control   Aphrodite Right Pre       No    3/5/25   15.6 198.5
2  4820 M     Control   Zeus      Right Pre       No    3/7/25   14.1 191.5
3  4823 M     Heat      Aphrodite Right Pre       No    3/20/25  14.4 239
4  4824 M     Heat      Apollo    Left  Pre       No    3/8/25   14.5 226.5
5  4825 M     Heat      Aphrodite Right Pre       No    3/9/25   13.3 168.5
6  4826 M     Control   Apollo    Left  Pre       No    3/11/25  14.8 192
# ℹ 25 more variables: Ketones <chr>, HCT <chr>, CORT <dbl>, Damage <dbl>,
#   X <lgl>, X.1 <lgl>, X.2 <lgl>, X.3 <lgl>, X.4 <lgl>, X.5 <lgl>, X.6 <lgl>,
#   X.7 <lgl>, X.8 <lgl>, X.9 <lgl>, X.10 <lgl>, SexF <fct>, TreatmentF <fct>,
#   TimepointF <fct>, N_group <int>, Mean_Damage_Group <dbl>,
#   SD_Damage_Group <dbl>, SE_Damage_Group <dbl>, CI_Margin_Damage <dbl>,
#   CI_Upper_Group <dbl>, CI_Lower_Group <dbl>
```

```
# 2. Cleaning up data frame before plotting
  # Define the consistent dodge width and rename the data frame for plotting
DODGE_WIDTH = 0.4
plot_data = datumlong2_with_summary

  #Order the time points so that Pre and Post are graphed in the correct order on the pl
plot_data$TreatmentF = as.factor(plot_data$TreatmentF)
plot_data$TimepointF = factor(plot_data$TimepointF, levels = c("Pre", "Post"))

# 3. Plotting the graph

plot = ggplot(data = plot_data, aes(x = TimepointF, y = Damage, color = TreatmentF)) +

  geom_jitter(
    aes(shape = SexF, group = TreatmentF),
    size = 3.5,
    alpha = 0.5,
    position = position_jitterdodge(
      jitter.width = 0.2,
      dodge.width = DODGE_WIDTH
  )
) +

  geom_point(
    aes(y = Mean_Damage_Group, group = TreatmentF),
    size = 7.5,
    shape = 18,
    stroke = 1,
    color = "black",
    position = position_dodge(width = DODGE_WIDTH)
  ) +

  geom_errorbar(
    aes(ymin = CI_Lower_Group, ymax = CI_Upper_Group, group = TreatmentF),
    width = 0.2,
    linewidth = 1,
    color = "black",
    position = position_dodge(width = DODGE_WIDTH)
  ) +

  geom_line(
    aes(y = Mean_Damage_Group, group = TreatmentF),
    linewidth = 1.5,
    position = position_dodge(width = DODGE_WIDTH)
  ) +

  scale_color_manual(
    values = c(
      "Control" = "#AAB8C2",
      "Heat" = "firebrick4"
    )
  ) +

  labs(
    x = "Time Point",
    y = "DNA Damage (AU)",
    color = "Treatment Group"
  ) +

  guides(shape = "none") +

  theme_classic() +


  theme(
    axis.title = element_text(face = "bold", size = 30),
    axis.text = element_text(size = 25),
    legend.justification = c(0.5, 0.5),
    legend.title = element_text(size = 28),
    legend.text = element_text(size = 25),
    strip.text = element_text(size = 20),
    panel.background = element_rect(fill = "transparent", color = NA),
    plot.background = element_rect(fill = "transparent", color = NA),
    legend.background = element_rect(fill = "transparent", color = NA),
    legend.box.background = element_rect(fill = "transparent", color = NA),
    axis.line = element_line(linewidth = 1.5, colour = "black")
  )

# 4. Adding signifigance and sample size

plot1 = plot +
  annotate(
    "text",
    x = c(.9, 1.1, 1.9, 2.1),
    y = c(2.1 + .3, 2.5 + .3, 3.1 + .3, 2 + .3),
    label = c("a", "a", "a", "a"),
    size = 10,
    color = "black") +

   # Add a new layer for the sample sizes
  annotate(
    "text",
    x = c(.9, 1.1, 1.9, 2.1),
    # Adjust the y-coordinates to be slightly lower
    y = c(2.1, 2.5, 3.1, 2),
    # Add your sample sizes here
    label = c("(34)", "(42)", "(36)", "(35)"),
    size = 7.5, # Make the font size slightly smaller
    color = "black")

# 5. Save the graph
ggsave(
  filename = "DamagePlotSex.png",
  plot = plot1,
  width = 11.04,
  height = 6.49,
  units = "in",
  dpi = 300,
  bg = "transparent"
)
```